Before we start, please connect to this wifi network

On your laptop please BTW

- SSID: `uocslub-wifi`
- Password: `screw OSI`

Also please make sure you have **putty** installed

The slide sources are available at github.com/uocsclub/zero-to-homerouter

This PDF is available at https://talks.uocsclub.net

From zero to home router

Exploring high-level concept of how routers work through a hands-on lab

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2026-02-26

Outline

Theory

What even is an IP?

Playing with layer 2

Building the router

Automating IPs

Connecting to the internet

Firewall

DISCLAIMER

Networking is very finicky and things can easily break

Futhermore we are using a very finicky piece of software for running the lab.

Things may only work for me, that's expected

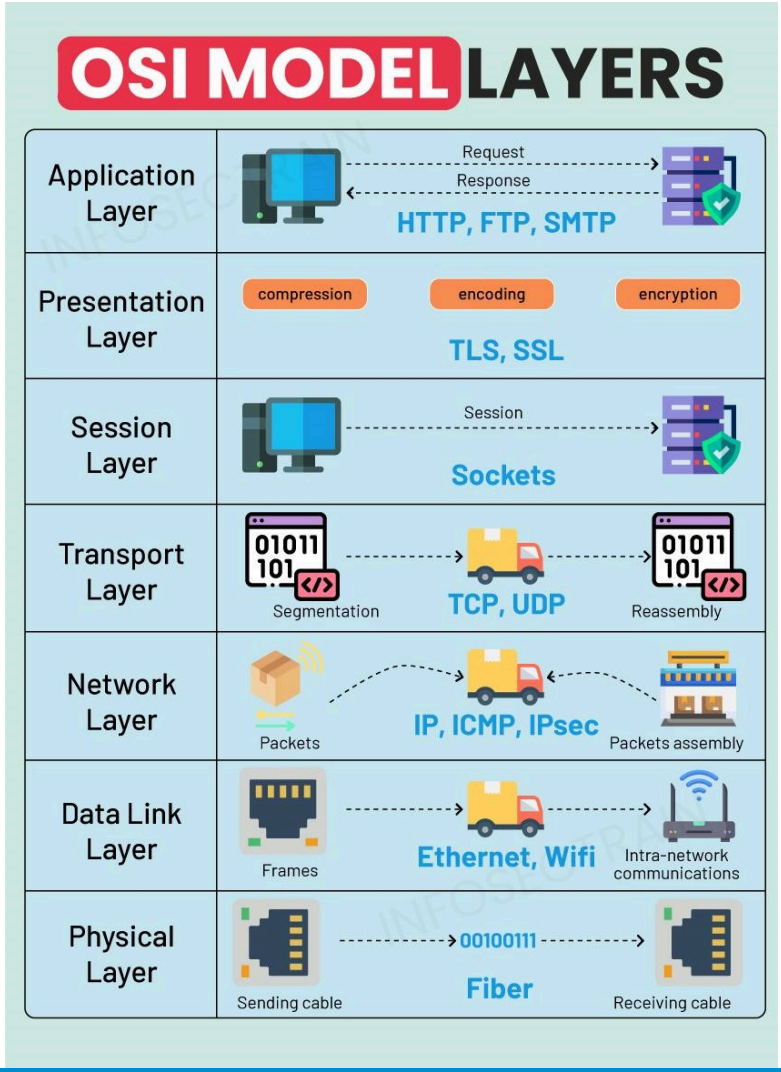
Things may break, you may get stuck, that's expected.

It is important that you let me know so I can get you unstuck

Let's get started!

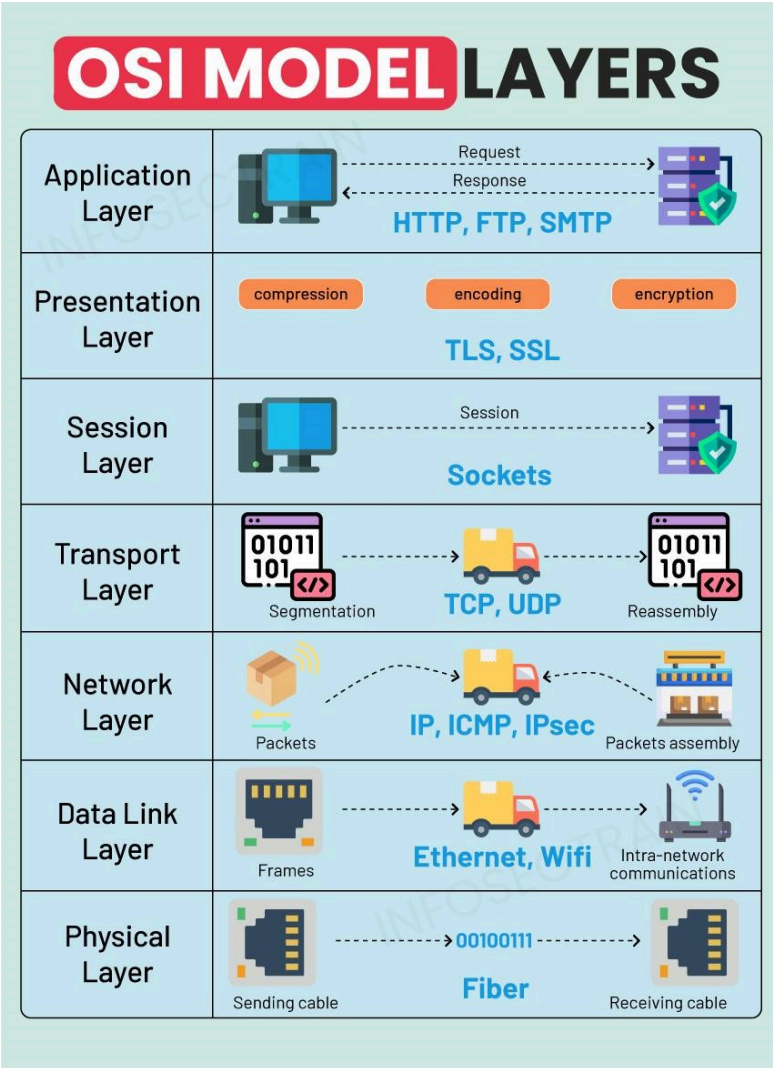
The dreaded OSI model

There are many criticisms of the OSI model, however understanding the first 3 layers is going to be crucial for this presentation



The Physical layer

The cable/fiber/Wireless signal you're using for communicating



The Data link layer

Yes, ethernet is not just the cable

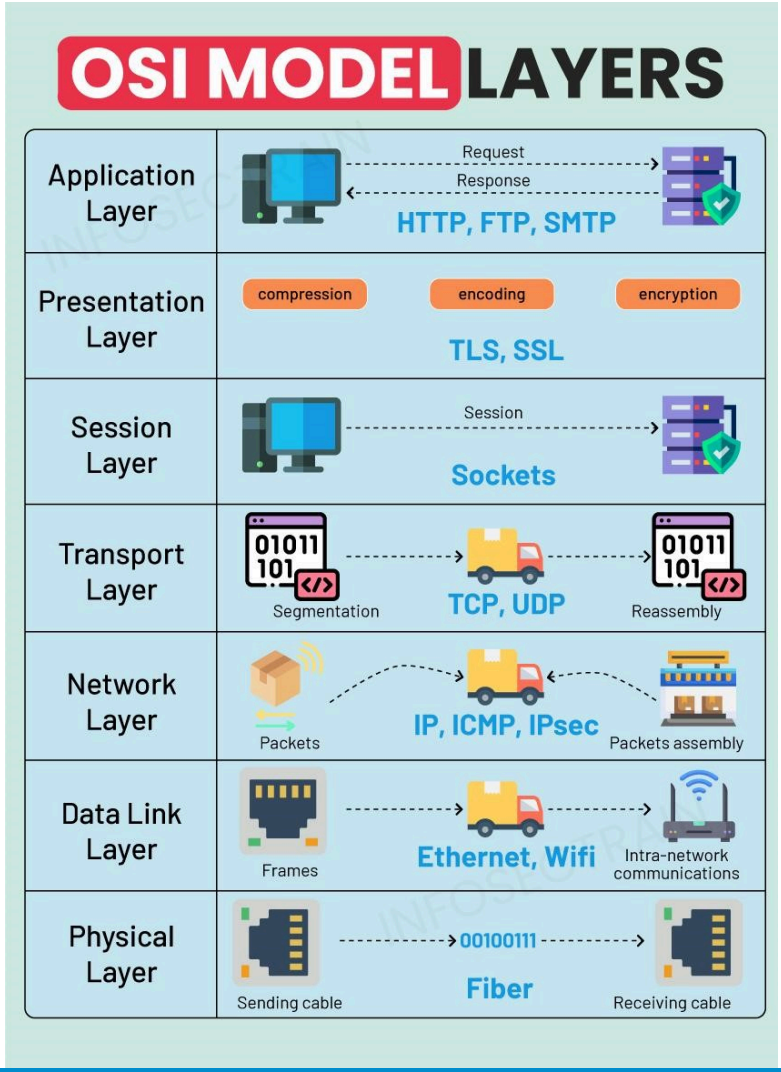
Ethernet dataframes operate on source and destination MAC addresses

Mac: Physical address of your network card, typically unchangable and manufacturer defined.

Format: `DE:AD:BE:EF:19:19`

Dataframe:

Preamble	SFD	Destination MAC Address	Source MAC Address	EtherType	Payload	FCS

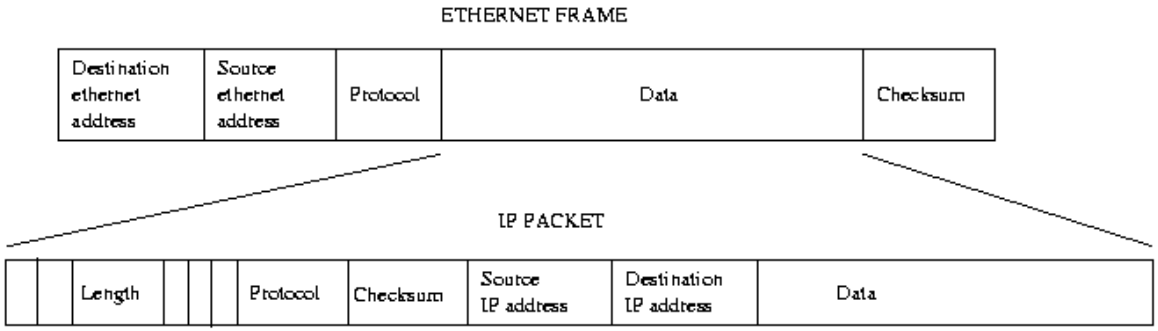


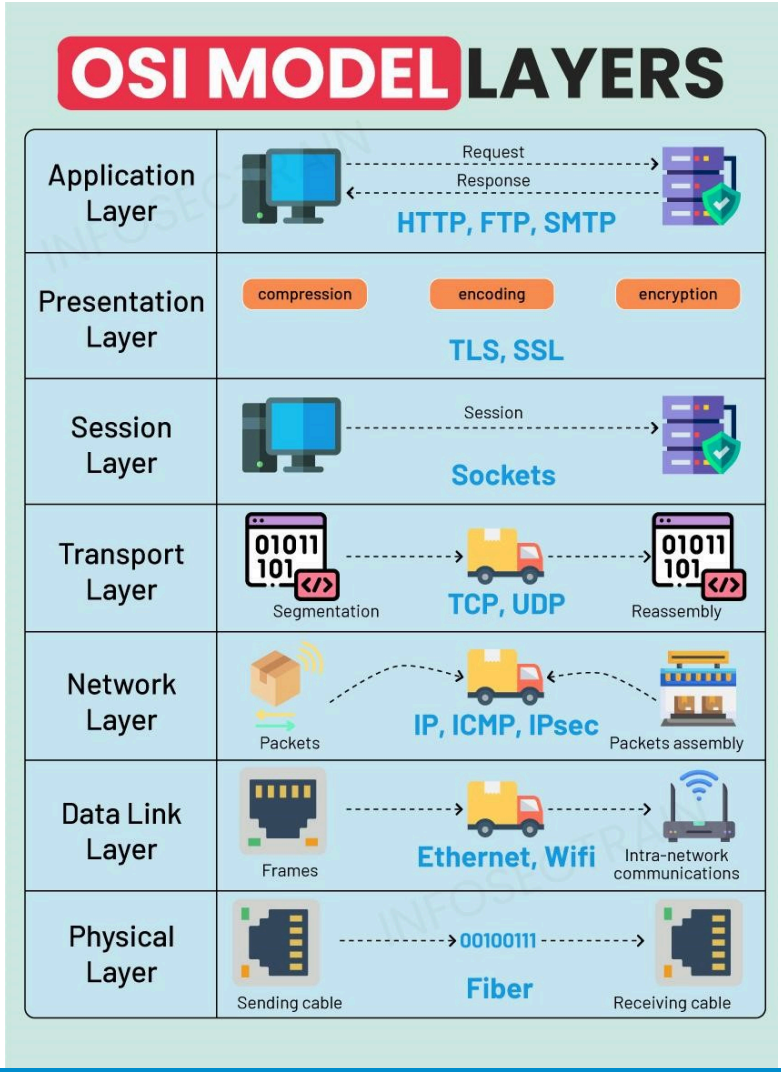
The Network layer

This is where your IP resides

This is where you get into different networks and being able to talk between those networks.

Layer 2 only deals with local transit, layer 3 allows you to talk to others

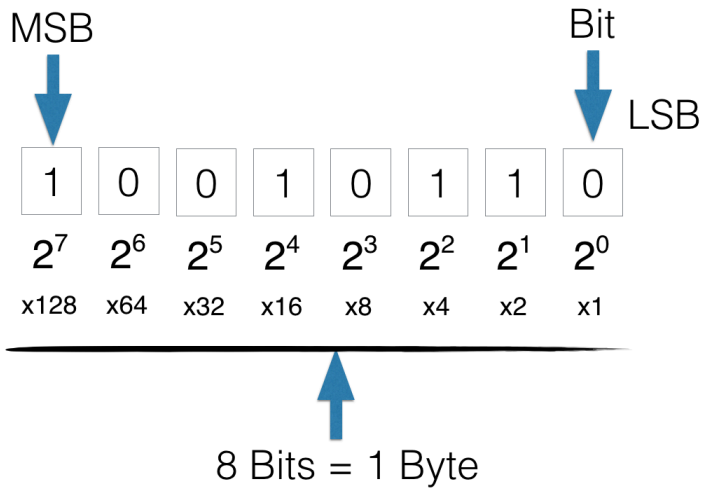




Understanding IPs

An ip is a representation of a 32bit integer, split into its 4 individual bytes:

32bit	11000000101010000000000000000001			
4x8bit	11000000	10101000	00000000	00000001
IP	192	168	0	1

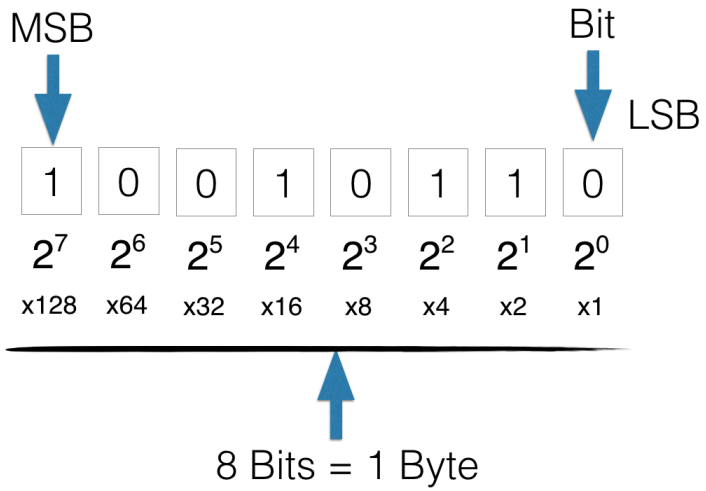


How IPs relate to eachother

You often see an ip written as the following `192.168.0.1/24`.

What is that /24?

IP	192	168	0	1
4x8bit	11000000	10101000	00000000	00000001
24bit subnet	11111111	11111111	11111111	00000000
Network	11000000	10101000	00000000	00000000
Network	192	168	0	0



Effective IP range of **192.168.0.1** to **192.168.0.254** (first and last are reserved)

Exercise

What is the network address for `192.168.3.19/16`?

Can it ping `192.168.10.3/16`?

How many useable addresses are on the network?

What is the subnet mask in IP form?

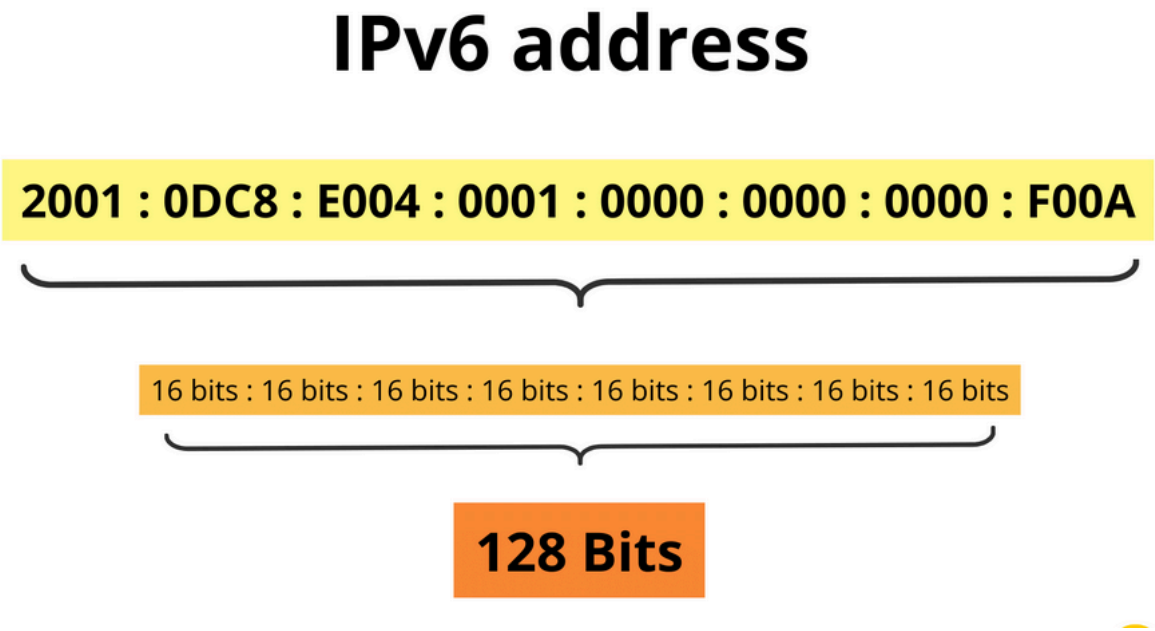
Answers

What is the network address for `192.168.3.19/16`?	**192.168.0.0/16**
Can it ping `192.168.10.3/16`?	**Yes it can ping**
How many useable addresses are on the network?	$(256 * 256) - 2$ **or** $2^{16} - 2$
What is the subnet mask in IP form?	**255.255.0.0**

Important ranges to memorize

IP	USE
127.0.0.0/8	Loopback addresses
192.168.0.0/16, 10.0.0.0/8, 172.16.0.0/12	Private ranges
100.64.0.0/10	For between ISPs and customers
0.0.0.0/8	Reserved for some local stuff
255.255.255.255/32	Reserved broadcast address

There are more ranges, however, the first 2 rows are the most important to know



Same thing concept as IPv4, but bigger and more annoying.

We won't go over them in this talk

Let's start labbing!

Again, ensure you're connected to the proper wifi

- SSID: `uocslab-wifi`
- Password: `screw OSI`

Then visit this ip: http://192.168.30.15:3080 (make sure it's not https)

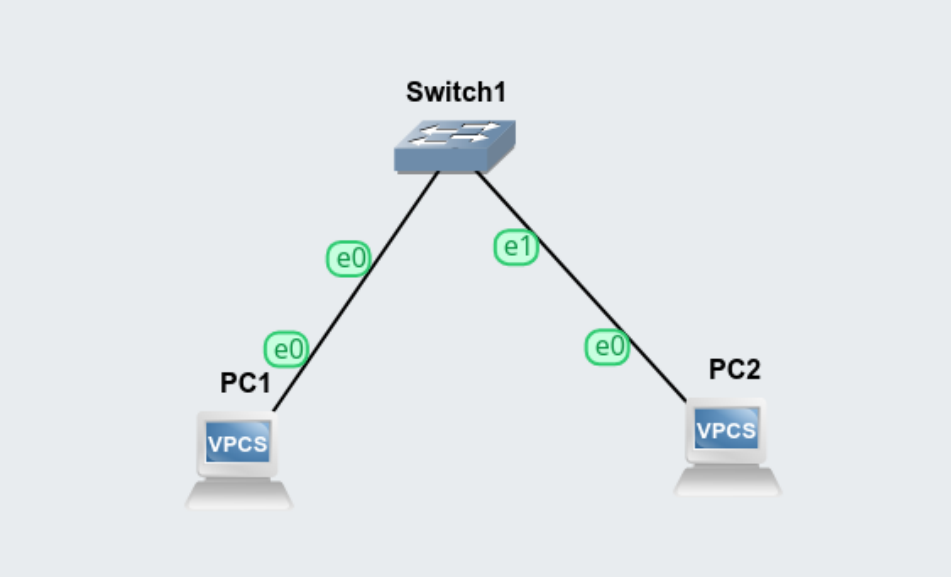
safari might cause issues, use firefox or chrome

New project

Create a new project and arrange devices in this configuration

The specific ethernet interfaces do not matter

You can then turn them on and we can test some stuff



Configuring the VPCs

VPC1

Open a web console on VPC1 and give it an ip

```
ip 192.168.0.1/24
```

```
PC2> ip 192.168.0.2/24
Checking for duplicate address...
PC2 : 192.168.0.2 255.255.255.0

PC2>
```

VPC2

Open a web console on VPC2 and give it an ip

```
ip 192.168.0.2/24
```

```
PC2> ip 192.168.0.2/24
Checking for duplicate address...
PC2 : 192.168.0.2 255.255.255.0

PC2>
```

Try pinging VPC2 from VPC1

Adding a 3rd VPC

Let's add a 3rd VPC outside the IP range and see how it behaves

VPC3

Open a web console on VPC3 and give it an ip

```
ip 192.168.1.1/24
```

```
PC3> ip 192.168.1.1/24
Checking for duplicate address...
PC3 : 192.168.1.1 255.255.255.0

PC3>
```

Try pinging VPC1 or VPC2 from VPC3

ARP?

VPC1

Take a look at the ARP table to see what's going on

```
show arp
```

```
PC1> show arp

00:50:79:66:68:00  192.168.0.2 expires in 109 seconds

PC1>
```

Presenter: *Clear the arp table, open wireshark and send your ping request again.*

Source	Destination	Protoc	Info
Private_66:68:01	Broadcast	ARP	Who has 192.168.0.2? Tell 192.168.0.1
Private_66:68:00	Private_66:68:01	ARP	192.168.0.2 is at 00:50:79:66:68:00
192.168.0.1	192.168.0.2	ICMP	Echo (ping) request id=0xae04, seq=1/25
192.168.0.2	192.168.0.1	ICMP	Echo (ping) reply id=0xae04, seq=1/25

What does this tell you?

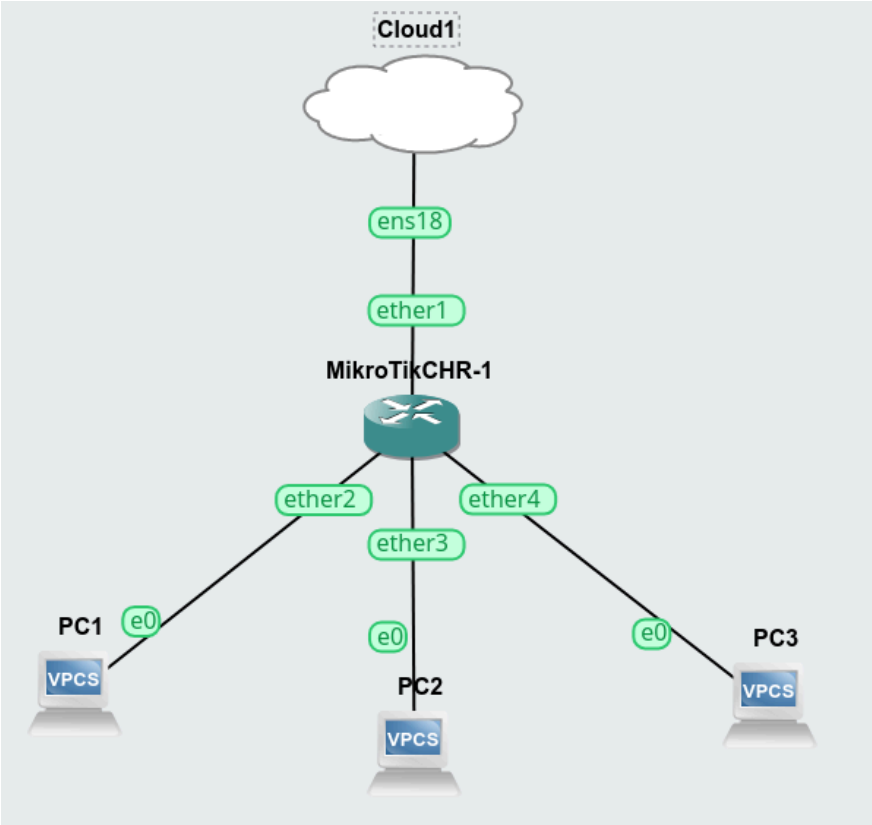
If it was a real switch, what would be in its ARP table?

Moving on to layer 3

Let's get rid of this cheap switch and install a nice shiny router (Mikrotik CHR).

Connect the VPCs to ports 2,3,4 and connect port 1 to a new device called "cloud"

Open up putty so we can connect to the new device

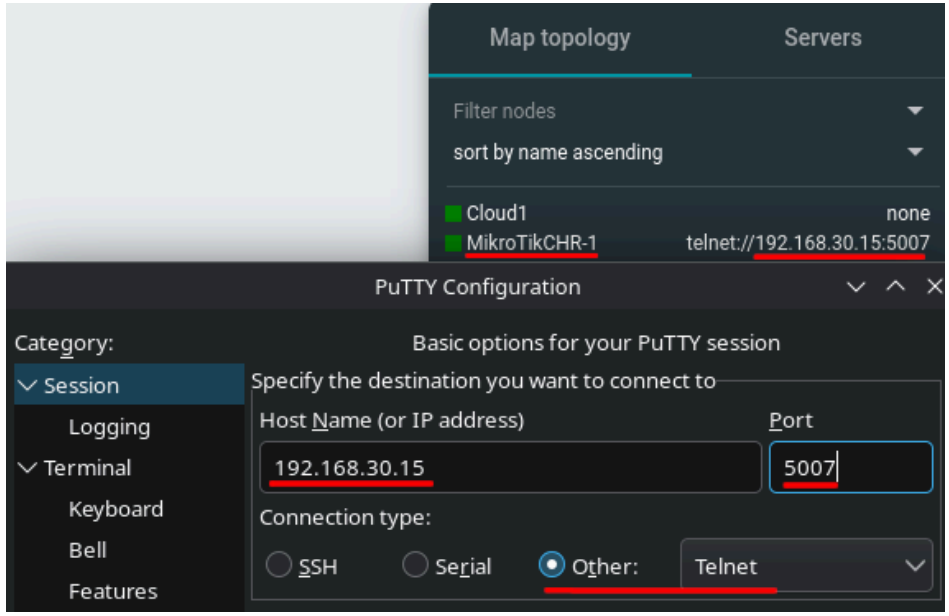


Connecting to the router

Look at the right of the GNS3 web interface, you will see a telnet IP and port for the router. Enter them into putty and connect.

- Username: admin
- No passowrd

You will need to accept the license, then ctrl+c the prompt to change the password



Removing the one pesky config it comes with

For some reason, the image comes with a DHCP client on ether1. We want that, but later, so we will remove it now so you can learn about it later.

You can remove it with `/ip/dhcp-client/remove ether1` (tab is your best friend here)

```
[admin@MikroTik] > /ip/dhcp-client/remove ether1
[admin@MikroTik] >
```

Let's get it talking to the VPCs

Try pinging from VPC1 to VPC2, see what happens.

Why is that?

Let’s add IPs on the router interface so the VPCs can talk

```
[admin@MikroTik] > /ip/address/add interface=ether2 address=192.168.0.101/24
[admin@MikroTik] > /ip/address/add interface=ether3 address=192.168.0.102/24
[admin@MikroTik] > /ip/address/add interface=ether4 address=192.168.1.101/24
```

Try pinging the router from VPC1, what about from VPC2. Try pinging between VPC1 and VPC2.

Why is that?

Presenter: *Open wireshark and see what happens on the other line*

Let's bring a switch back

Let’s remove the ip addresses on ether 2 and 3, and create a bridge.

We will then add these interfaces to the bridge.

```
[admin@MikroTik] /ip/arp> /ip/address/print
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS            NETWORK      INTERFACE
0 192.168.0.101/24   192.168.0.0  ether2
1 192.168.0.102/24   192.168.0.0  ether3
2 192.168.1.101/24   192.168.1.0  ether4
[admin@MikroTik] /ip/arp> /ip/address/remove numbers=0,1
[admin@MikroTik] /ip/arp> /interface/bridge/add name=bridge1
[admin@MikroTik] /ip/arp> /interface/bridge/port/add bridge=bridge1 interface=ether2
[admin@MikroTik] /ip/arp> /interface/bridge/port/add bridge=bridge1 interface=ether3
[admin@MikroTik] /ip/arp>
```

Can VPC1 ping VPC2?

You might have noticed we are lacking an ip on our bridge, let's solve that now.

```
[admin@MikroTik] > /ip/address/add interface=bridge1 address=192.168.0.101/24
[admin@MikroTik] >
```

Now can VPC1 and VPC2 ping the router?

What about pinging VPC3? What do you think is missing?

Is this a VPC or router issue?

Let's fix the VPC

VPCs 1 and 2 cannot talk to VPC3 since they are on different networks

That is exactly what a router is for, “routing” from one network to another

Let’s tell our VPC1 to use it to access other networks by setting the gateway (default route)

Does this fix it?

```
ip 192.168.0.1/24 192.168.0.101
```

```
PC1> ip 192.168.0.1/24 192.168.0.101
Checking for duplicate address...
PC1 : 192.168.0.1 255.255.255.0 gateway 192.168.0.101

PC1>
```

Getting our first inter-network communication

Let's add the missing gateway in VPC3 so we can finally talk between networks

Finally now, does everything work?

```
ip 192.168.1.1/24 192.168.1.101
```

```
PC3> ip 192.168.1.1 192.168.1.101
Checking for duplicate address...
PC3 : 192.168.1.1 255.255.255.0 gateway 192.168.1.101

PC3>
```

Wow that was tedious

That was so tedious though…

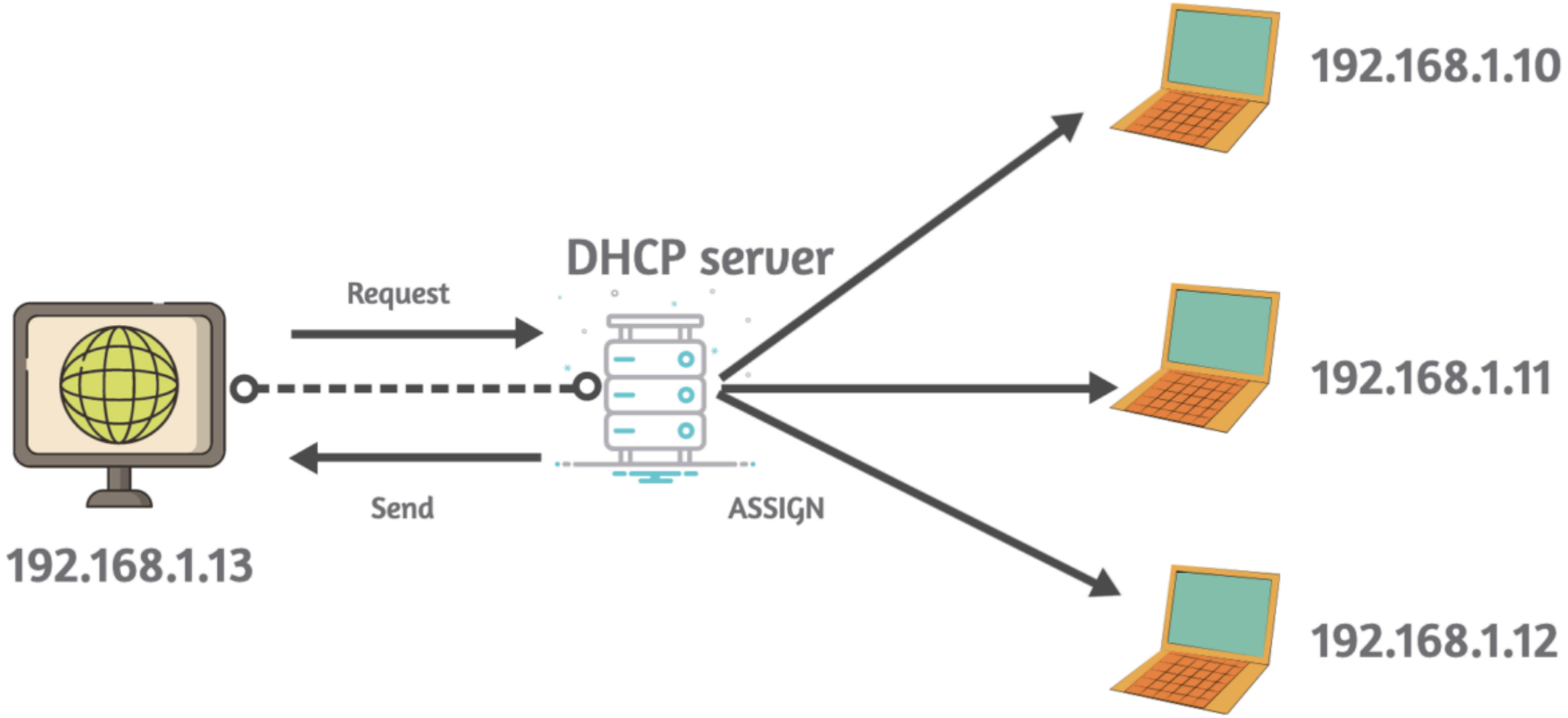
I don't have to do this on my computer.

Surely there is some way to automate this

Maybe there is some sort of

Dynamic **H**ost **C**onfiguration **P**rotocol we can use…

Oh wow... How convenient



Let's setup DHCP

DHCP has a client/server model.

There is a server provisioning IPs, and the clients run on your devices and receive them.

Let’s create a server on the router so our VPCs can get IPs

We will leave VPC3 with a static IP

`/ip/dhcp-server-setup`

```
[admin@MikroTik] > /ip/dhcp-server/setup
Select interface to run DHCP server on

dhcp server interface: bridge1
Select network for DHCP addresses

dhcp address space: 192.168.0.0/24
Select gateway for given network

gateway for dhcp network: 192.168.0.101
Select pool of ip addresses given out by DHCP server

addresses to give out: 192.168.0.1-192.168.0.100
Select DNS servers

dns servers:
Select lease time

lease time: 1800
[admin@MikroTik] >
```

Obtaining IPs

We can easily get dhcp provisioning on our VPCs using the `dhcp` command

If we see DORA we are good, if it freezes halfway, we have some debugging to do

Let's make sure everyone still pings eachother

`dhcp`

```
PC1> dhcp
DORA IP 192.168.0.99/24 GW 192.168.0.101

PC1>
```

`dhcp`

```
PC2> dhcp
DORA IP 192.168.0.100/24 GW 192.168.0.101

PC2>
```

What about that DHCP-client we deleted?

You are correct, we will need to bring that back.

You now know what it does so we are allowed to use it.

We can add it back and then see that we have received an IP.

This IP comes from DHCP server on the router you're all connected to for this lab!

This means you can ping it from your own laptop.

```
/ip/dhcp-client/add interface=ether1
```

```
[admin@MikroTik] > /ip/dhcp-client/add interface=ether1
[admin@MikroTik] > /ip/address/print
Flags: D - DYNAMIC
Columns: ADDRESS, NETWORK, INTERFACE
#   ADDRESS            NETWORK       INTERFACE
0   192.168.1.101/24   192.168.1.0   ether4
1   192.168.0.101/24   192.168.0.0   bridge1
2 D 192.168.29.244/22  192.168.28.0  ether1
[admin@MikroTik] >
```

```
ping <ip>
```

```
bash-5.3$ ping -c 2 192.168.29.244
PING 192.168.29.244 (192.168.29.244) 56(84) bytes of data.
64 bytes from 192.168.29.244: icmp_seq=1 ttl=64 time=12.8 ms
64 bytes from 192.168.29.244: icmp_seq=2 ttl=64 time=28.8 ms

--- 192.168.29.244 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 12.841/20.798/28.756/7.957 ms
bash-5.3$
```

Do we have internet now?

Let's try!

Let's see if our virtual router can ping our physical router?

Can it ping 1.1.1.1 (cloudflare)?

```
ping 192.168.30.1
```

```
[admin@MikroTik] > ping 192.168.30.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 192.168.30.1                               56  64 576us
    1 192.168.30.1                               56  64 568us
    2 192.168.30.1                               56  64 647us
    sent=3 received=3 packet-loss=0% min-rtt=568us avg-rtt=597us
   max-rtt=647us

[admin@MikroTik] >
```

```
ping 1.1.1.1
```

```
[admin@MikroTik] > ping 1.1.1.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 1.1.1.1                                    56  52 22ms783us
    1 1.1.1.1                                    56  52 24ms134us
    2 1.1.1.1                                    56  52 23ms342us
    sent=3 received=3 packet-loss=0% min-rtt=22ms783us avg-rtt=23ms419us
   max-rtt=24ms134us

[admin@MikroTik] >
```

How does the router know where to route stuff

Let's take a look at the routing table and see where packets get sent

We can see the entries it discovered from the ip addresses on the interfaces

We can also see the gateway (`0.0.0.0`) that was populated from the dhcp-client

`/ip/route/print`

```
[admin@MikroTik] > /ip/route/print
Flags: D - DYNAMIC; A - ACTIVE; c - CONNECT, d - DHCP
Columns: DST-ADDRESS, GATEWAY, DISTANCE
    DST-ADDRESS       GATEWAY        DISTANCE
DAd 0.0.0.0/0         192.168.30.1          1
DAc 192.168.0.0/24    bridge1               0
DAc 192.168.1.0/24    ether4                0
DAc 192.168.28.0/22   ether1                0
[admin@MikroTik] >
```

Can our clients ping that stuff?

Let's try!

Let's see if our VPC can ping our physical router?

Can it ping 1.1.1.1 (cloudflare)?

uhmm

`ping 192.168.30.1`

```
PC2> ping 192.168.30.1

192.168.30.1 icmp_seq=1 timeout
192.168.30.1 icmp_seq=2 timeout
^C
PC2>
```

`ping 1.1.1.1`

```
PC2> ping 1.1.1.1

1.1.1.1 icmp_seq=1 timeout
1.1.1.1 icmp_seq=2 timeout
^C
PC2>
```

Maybe things are not quite so simple

Presenter, open wireshark

Let's try pinging again, see what goes on the network!

Source	Destination	Protoc	Info
192.168.0.100	192.168.30.1	ICMP	Echo (ping) request id=0xf60f, seq=1/256, ttl=63 (no response found!)
192.168.0.100	192.168.30.1	ICMP	Echo (ping) request id=0xf80f, seq=2/512, ttl=63 (no response found!)

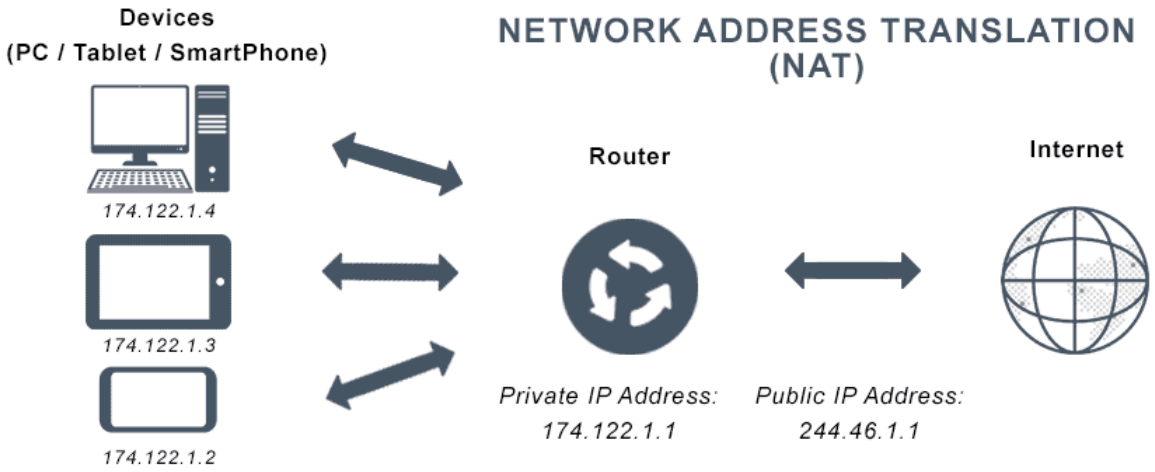
So there definitely are packets going out to the real router, but nothing coming back...

Why is that?

The answer is NAT

The real router doesn't know where to find `192.168.0.0/24`, so it sends the packet to it's own gateway, instead of back to our virtual router

NAT fixes this by allowing us to hide IPs behind a public facing one (among other things)



Simple one line fix

Let's type the magic line that fixes it for our router

The action is called "IP masquerading", where our virtual router puts its IP as the source IP so it can send the packet on our behalf, then it will send it back to us by fixing the proper destination address on the reply

```
/ip/firewall/nat/add chain=srcnat action=masquerade
```

```
[admin@MikroTik] > /ip/firewall/nat/add chain=srcnat action=masquerade
[admin@MikroTik] > /ip/firewall/nat/print
Flags: X - disabled, I - invalid; D - dynamic
 0    chain=srcnat action=masquerade
[admin@MikroTik] >
```

Third times the charm

```
PC1> ping 192.168.30.1

84 bytes from 192.168.30.1 icmp_seq=1 ttl=63 time=1.135 ms
84 bytes from 192.168.30.1 icmp_seq=2 ttl=63 time=1.370 ms
^C
PC1> ping 1.1.1.1

84 bytes from 1.1.1.1 icmp_seq=1 ttl=51 time=24.386 ms
84 bytes from 1.1.1.1 icmp_seq=2 ttl=51 time=25.567 ms
^C
PC1>
```

Nice

Is it always necessary

```
sudo ip route add 192.168.0.0/24 via
<router-ip>
```

No, if your device knows where to go, it should just work

Presenter, add a static route on your laptop

The presenter should now be able to ping the VPCs

```
bash-5.3$ ping 192.168.0.100
PING 192.168.0.100 (192.168.0.100) 56(84) bytes of data.
^C
--- 192.168.0.100 ping statistics ---
2 packets transmitted, 0 received, 100% packet loss, time 1038ms

bash-5.3$ sudo ip route add 192.168.0.0/24 via 192.168.29.244
bash-5.3$ ping 192.168.0.100
PING 192.168.0.100 (192.168.0.100) 56(84) bytes of data.
64 bytes from 192.168.0.100: icmp_seq=1 ttl=63 time=8.24 ms
64 bytes from 192.168.0.100: icmp_seq=2 ttl=63 time=4.21 ms
64 bytes from 192.168.0.100: icmp_seq=3 ttl=63 time=2.74 ms
^C
--- 192.168.0.100 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 2.743/5.063/8.242/2.325 ms
bash-5.3$
```

What was that... a route?

We saw the concept of a default route (gateway).

There also exists a more general concept of routes where you indicate how to reach specific networks

We can see the routing table in our main router like this

Routes can be set manually or automatically

`/ip/route/print`

```
[admin@MikroTik] > /ip/route/print
Flags: D - DYNAMIC; A - ACTIVE; c - CONNECT, d - DHCP
Columns: DST-ADDRESS, GATEWAY, DISTANCE
    DST-ADDRESS       GATEWAY        DISTANCE
DAd 0.0.0.0/0         192.168.30.1          1
DAc 192.168.0.0/24    bridge1               0
DAc 192.168.1.0/24    ether4                0
DAc 192.168.28.0/22   ether1                0
[admin@MikroTik] >
```

Dynamic routing

There are more advanced automatic routing protocols such as **OSPF** or **BGP**

Those are beyond the scope of this presentation.

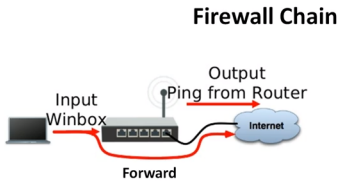
Fun fact: **BGP** caused the 2021 Facebook outage

But I don't want my network to be open

You don’t want just anyone to be able to access the devices on your home network

To get some form of isolation, you will need a firewall

Let’s go through the important default rules that might ship with your router



Rule 1

```
add action=accept chain=input connection-state=established,related,untracked
```

This allows existing connections to keep talking on the open socket.

This is important because later we will block incoming connections

Rule 2

```
add action=drop chain=input connection-state=invalid
```

You don't want to handle input to closed sockets, you should drop those

Rule 3

```
add action=accept chain=input protocol=icmp
```

Allow the router to be pinged. This is useful if you want to know if your device is up.

Some people prefer to have this off (cough* microsoft)

This is necessary since again, we'll turn off input later

Rule 4

```
add action=drop chain=input in-interface=!bridge1
```

Block the input chain to users on the local network only

This prevents external bad actors from trying to login and change settings

This should disallow your laptop from accessing the routers web interface

Rules 5 and 6

```
add action=accept chain=forward connection-state=established,related,untracked
```

```
add action=drop chain=forward connection-state=invalid
```

Same as rules 1 and 2 but for forwarding.

This is because we're going to once again restrict traffic to local only

Rule 7

```
add action=drop chain=forward connection-state=new in-interface=!bridge1
                                connection-nat-state=!dstnat
```

This stops external devices from trying to use the router to route to local devices

Now the static route demo should no longer work

That final filter is to allow port-forwarding (which is a form of NAT btw)

We made it

That was a lot, but we made it!

We now understand the fundamental inner-workings of your typical home router.

Recap

We started with **MAC** addresses, **IPs** how they are split into networks

Next, we saw basic how **switches** work and **ARP** tables

Next, we went over how **routers** bridge networks

We also saw how **routing tables** work

We then saw how to dynamically dispatch IPs with **DHCP**

Then we saw how to hide ourselves behind a **public facing IP**

Finally we setup a basic **firewall** to isolate our LAN